10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

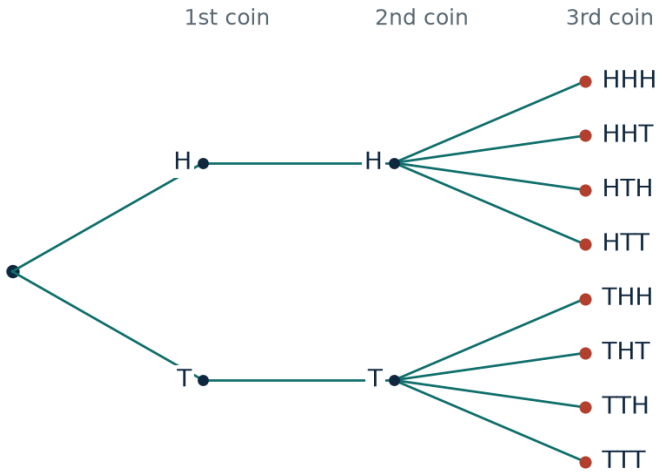
Exercise 12.1 — Probability (Sample Space and Simple Probability)

Unit 12 | Exercise 12.1 | New Book (PECTAA)

Exercise 12.1 (Solved)

In this exercise we only write the sample space and then count. Remember: P(E) = n(E) / n(S), and the probability of every event lies between 0 and 1.

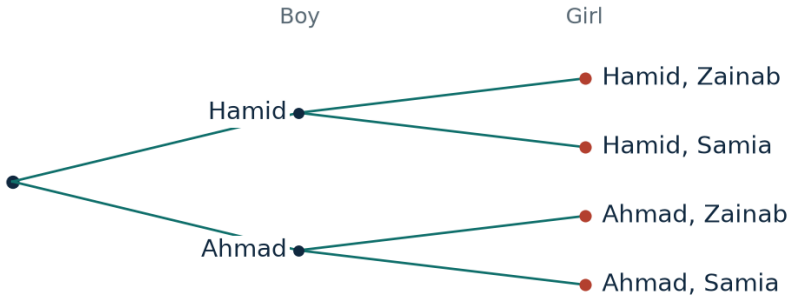
1. Find the sample space for tossing three coins using tree diagram.



Tree diagram of tossing three coins

Answer: S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, n(S) = 8

2. A teacher randomly selects one boy and one girl from a group of 2 boys (Hamid, Ahmad) and 2 girls (Zainab, Samia). Draw a tree diagram and list the sample space for all possible outcomes.



Tree diagram for selecting one boy and one girl

Answer: S = {Hamid-Zainab, Hamid-Samia, Ahmad-Zainab, Ahmad-Samia}, n(S) = 4

3. If A is an event of a random experiment such that P(A) : P(Ā) = 17 : 15 and n(S) = 640, then find (i) P(Ā) (ii) n(A).

We know that P(A) + P(Ā) = 1.

Let P(A) = 17k and P(Ā) = 15k.

17k + 15k = 1 ⇒ 32k = 1 ⇒ k = 1 / 32

(i) P(Ā) = 15k = 15 / 32

(ii) P(A) = n(A) / n(S) ⇒ 17 / 32 = n(A) / 640

n(A) = 17 × 640 / 32 = 17 × 20 = 340

Answer: (i) P(Ā) = 15/32 (ii) n(A) = 340

4. A coin is tossed thrice. What is the probability of getting two consecutive tails?

When a coin is tossed thrice, the sample space is

S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, n(S) = 8

Let A be the event of getting two consecutive tails.

A = {HTT, TTH, TTT}, n(A) = 3

P(A) = n(A) / n(S) = 3 / 8

Answer: P(two consecutive tails) = 3/8

5. A dice is rolled and coin is tossed together. (i) Find sample space by drawing possibility diagram. (ii) Find sample space by sketching tree diagram. (iii) What is probability of getting a tail and an even number?

(i) Possibility diagram:

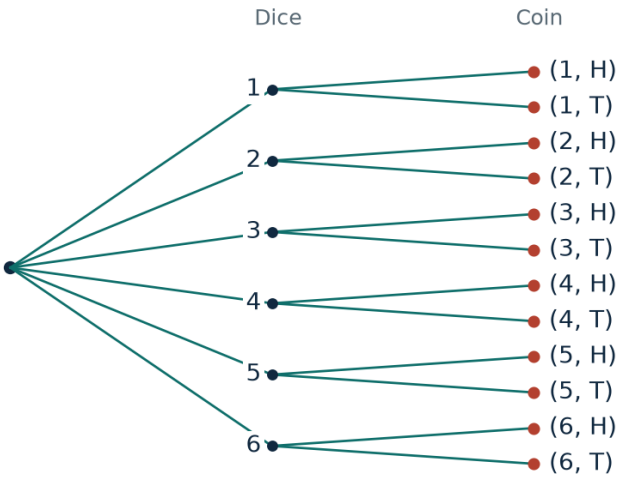
Coin **Dice**

	1	2	3	4	5	6
H	(H, 1)	(H, 2)	(H, 3)	(H, 4)	(H, 5)	(H, 6)
T	(T, 1)	(T, 2)	(T, 3)	(T, 4)	(T, 5)	(T, 6)

Possibility diagram of one coin and one dice

S = {(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)}, n(S) = 12

(ii) Tree diagram:



Tree diagram of one dice and one coin

(iii) Let E be the event of getting a tail and an even number.

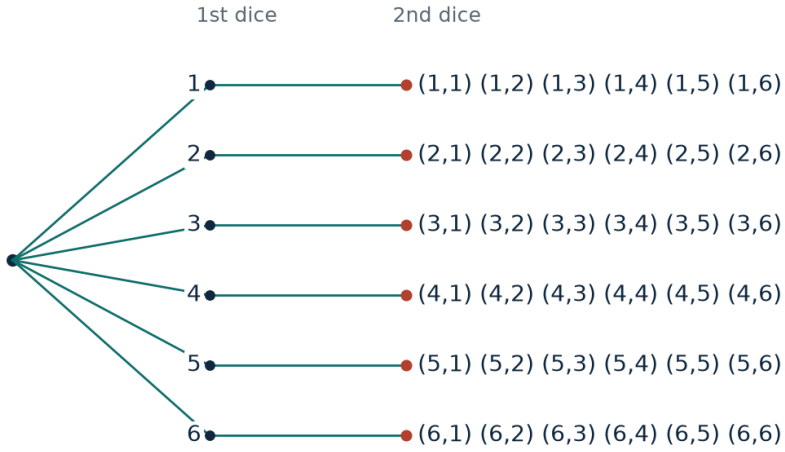
E = {(T, 2), (T, 4), (T, 6)}, n(E) = 3

P(E) = n(E) / n(S) = 3 / 12 = 1 / 4

Answer: (i) and (ii) Sample space written above (iii) P(T and even) = 1/4

6. Two unbiased dice are rolled once. (i) Find sample space by sketching tree diagram. (ii) Find sample space by drawing possibility diagram. (iii) Find the probability of getting (a) same number on both dice (b) the product as a prime number (c) the sum as an even number (d) the sum as 13.

(i) Tree diagram:



Tree diagram of rolling two dice

(ii) Possibility diagram:

1st dice **2nd dice**

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Possibility diagram of rolling two dice

From the possibility diagram n(S) = 36.

(iii) (a) Same number on both dice:

A = {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)}, n(A) = 6

P(A) = 6 / 36 = 1 / 6

(b) Product as a prime number:

B = {(1, 2), (1, 3), (1, 5), (2, 1), (3, 1), (5, 1)}, n(B) = 6

P(B) = 6 / 36 = 1 / 6

(c) Sum as an even number:

C = {(1, 1), (1, 3), (1, 5), (2, 2), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5),
(4, 2), (4, 4), (4, 6), (5, 1), (5, 3), (5, 5), (6, 2), (6, 4), (6, 6)}

n(C) = 18

P(C) = 18 / 36 = 1 / 2

(d) Sum as 13:

The largest sum on two dice is 6 + 6 = 12, so a sum of 13 is impossible.

P(sum 13) = 0 / 36 = 0

Answer: (iii) (a) 1/6 (b) 1/6 (c) 1/2 (d) 0

7. Three fair coins are tossed together. Find the probability of getting (i) all tails (ii) at least one head (iii) at most two tails (iv) 2 heads (v) at most 2 heads (vi) no head.

Here S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, n(S) = 8

(i) A = {TTT}, n(A) = 1 ⇒ P(A) = 1 / 8

(ii) B = all outcomes except TTT, n(B) = 7 ⇒ P(B) = 7 / 8

(iii) C = outcomes with 0, 1 or 2 tails = all except TTT, n(C) = 7

P(C) = 7 / 8

(iv) D = {HHT, HTH, THH}, n(D) = 3 ⇒ P(D) = 3 / 8

(v) E = all except HHH, n(E) = 7 ⇒ P(E) = 7 / 8

(vi) F = {TTT}, n(F) = 1 ⇒ P(F) = 1 / 8

Answer: (i) 1/8 (ii) 7/8 (iii) 7/8 (iv) 3/8 (v) 7/8 (vi) 1/8

8. A bag contains 4 red balls, 5 white balls, 6 green balls and 3 black balls. Ali draws a ball at random from the bag. Find the probability that the ball drawn is (i) white (ii) red (iii) not white (iv) not black.

Total number of balls n(S) = 4 + 5 + 6 + 3 = 18

(i) White balls n(W) = 5 ⇒ P(W) = 5 / 18

(ii) Red balls n(R) = 4 ⇒ P(R) = 4 / 18 = 2 / 9

(iii) P(not white) = 1 − P(W) = 1 − 5 / 18 = 13 / 18

(iv) P(not black) = 1 − 3 / 18 = 15 / 18 = 5 / 6

Answer: (i) 5/18 (ii) 2/9 (iii) 13/18 (iv) 5/6

9. A number is selected at random from the set of whole numbers 1 to 15, both inclusive. Find the probability that the number selected is: (i) odd (ii) a multiple of 5 (iii) the square of 2 (iv) prime (v) 20.

S = {1, 2, 3, ..., 15}, n(S) = 15

(i) Odd numbers A = {1, 3, 5, 7, 9, 11, 13, 15}, n(A) = 8

P(A) = 8 / 15

(ii) Multiples of 5, B = {5, 10, 15}, n(B) = 3 ⇒ P(B) = 3 / 15 = 1 / 5

(iii) Square of 2, C = {4}, n(C) = 1 ⇒ P(C) = 1 / 15

(iv) Primes D = {2, 3, 5, 7, 11, 13}, n(D) = 6 ⇒ P(D) = 6 / 15 = 2 / 5

(v) 20 is not in S, so E = { } and P(E) = 0

Answer: (i) 8/15 (ii) 1/5 (iii) 1/15 (iv) 2/5 (v) 0

10. If the probability of an event A is 7/10, then find the probability of the event “not A”.

P(A) = 7 / 10

P(not A) = 1 − P(A) = 1 − 7 / 10 = 3 / 10

Answer: P(not A) = 3/10

11. A dice is rolled twice. Find the probability of having a number greater than 4 on each roll.

n(S) = 6 × 6 = 36

A number greater than 4 means 5 or 6, so on one roll there are 2 outcomes.

A = {(5, 5), (5, 6), (6, 5), (6, 6)}, n(A) = 4

P(A) = 4 / 36 = 1 / 9

Answer: P(greater than 4 on each roll) = 1/9

Remember

2 coins → n(S) = 4, 3 coins → 8, two dice → 36.

P(not A) = 1 − P(A) is often the shortest way.